36 Resource Acquisition and Transport in Vascular Plants

KEY CONCEPTS

36.1 Adaptations for acquiring resources were key steps in the evolution of vascular plants *p. 785*

36.2 Different mechanisms transport substances over short or long distances *p. 787*

36.3 Transpiration drives the transport of water and minerals from roots to shoots via the xylem *p. 792*

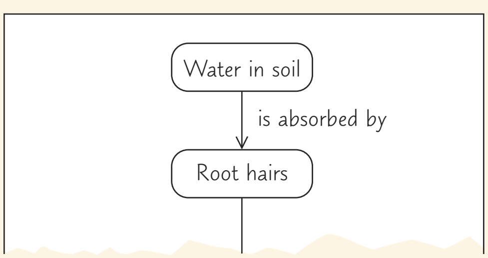
36.4 The rate of transpiration is regulated by stomata *p. 796*

36.5 Sugars are transported from sources to sinks via the phloem *p. 799*

36.6 The symplast is highly dynamic *p. 801*

Study Tip

Make a flowchart: Make a simple flowchart of the movement of water molecules from the soil through the tissues of a plant to the outside air.



Go to Mastering Biology

For Students (in eText and Study Area)
- Get Ready for Chapter 36
- Figure 36.9 Walkthrough: Transport of Water and Minerals from Root Hairs to the Xylem
- BioFlix® Animation: Water Transport in Plants

For Instructors to Assign (in Item Library)
- BioFlix® Tutorial: Water Transport in Plants: The Transpiration-Cohesion-Tension Mechanism
- BioFlix® Tutorial: Transpiration

Ready-to-Go Teaching Module (in Instructor Resources)
- Transport in Plants (Concept 36.2)

Figure 36.1 This English ivy has covered every square centimeter of a wall with foliage, an effective way to acquire light energy for photosynthesis. The sugars produced by the leaves and the water and minerals absorbed by the roots have to be transported throughout the plant.

What causes the movement of water, minerals, and sugars in most vascular plants?

Water and minerals are pulled up from the roots by **negative pressure** (tension) generated by evaporation from the leaves.

Sugars are pushed by **positive pressure** from where they are produced or stored to where they are needed. They can move both ways between leaves and roots.